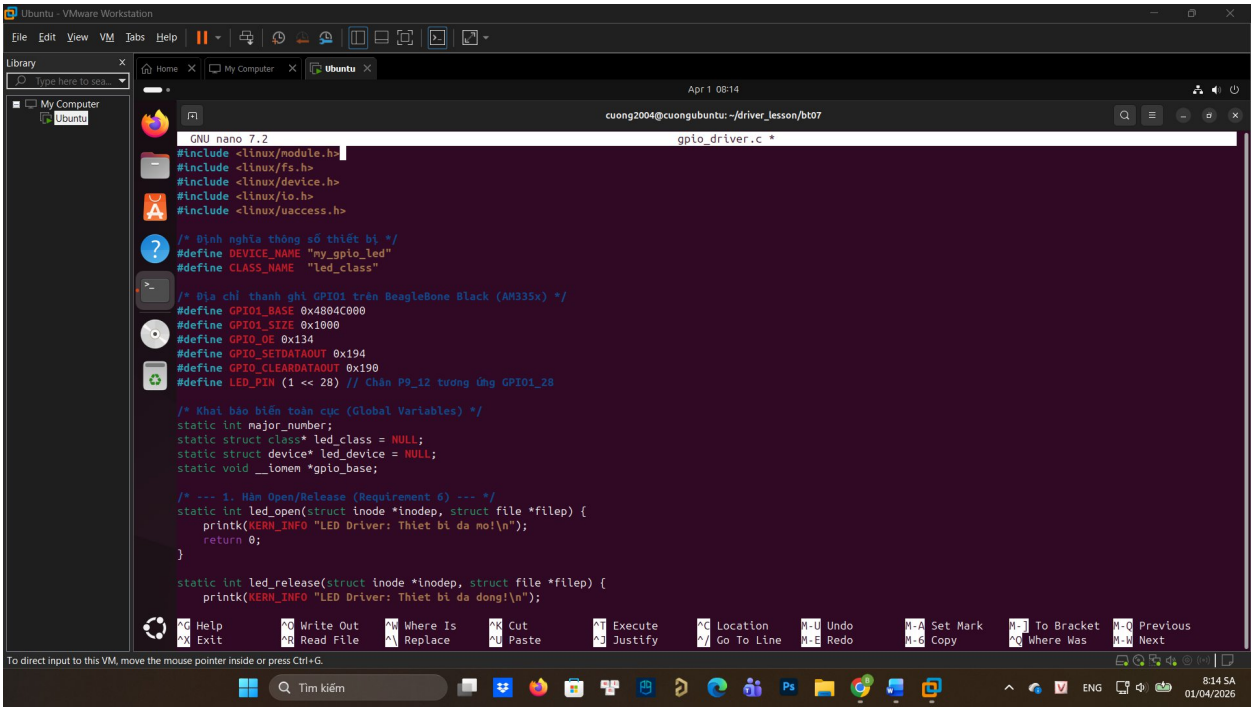
tạo một không gian riêng để chứa Driver

```
cuong2004@cuongubuntu:~$ mkdir -p ~/driver_lesson/bt07
cd ~/driver_lesson/bt07
cuong2004@cuongubuntu:~/driver_lesson/bt07$
```

Viết mã nguồn Driver (gpio_driver.c)



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cuong2004@cuongubuntu: ~/driver_lesson/bt07

GNU nano 7.2 gpio_driver.c *

```
}

static int led_release(struct inode *inodep, struct file *filep) {
    printk(KERN_INFO "LED Driver: Thiet bi da dong!\n");
    return 0;
}

/* --- 2. Hàm Read (Requirement 7, 22) --- */
static ssize_t led_read(struct file *filep, char *buffer, size_t len, loff_t *offset) {
    char status = '0';
    // Logic: Có thể đọc trạng thái từ thanh ghi DATAIN ở đây
    if (copy_to_user(buffer, &status, 1)) return -EFAULT;
    return 1;
}

/* --- 3. Hàm Write (Requirement 7, 19) --- */
static ssize_t led_write(struct file *filep, const char *buffer, size_t len, loff_t *offset) {
    char val;
    if (copy_from_user(&val, buffer, 1)) return -EFAULT;

    if (val == '1') {
        iowrite32(LED_PIN, gpio_base + GPIO_SETDATAOUT); // Bật LED
        printk(KERN_INFO "LED Driver: LED ON\n");
    } else if (val == '0') {
        iowrite32(LED_PIN, gpio_base + GPIO_CLEARDATAOUT); // Tắt LED
        printk(KERN_INFO "LED Driver: LED OFF\n");
    }
    return len;
}

/* --- 4. Đăng ký file_operations (Requirement 10) --- */
static struct file_operations fops = {
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous
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cuong2004@cuongubuntu: ~/driver_lesson/bt07

GNU nano 7.2 gpio_driver.c *

```
/* --- 4. Đăng ký file_operations (Requirement 10) --- */
static struct file_operations fops = {
    .owner = THIS_MODULE,
    .open = led_open,
    .read = led_read,
    .write = led_write,
    .release = led_release,
};

/* --- 5. Hàm Khởi tạo - Init (Requirement 4, 11, 15, 16, 24) --- */
static int __init led_init(void) {
    uint32_t oe_val; // Khai báo biến lên đầu hàm để tránh lỗi C90

    printk(KERN_INFO "LED Driver: Dang bat dau khoi tao...\n");

    // Tự động cấp phát Major/Minor (Requirement 11)
    major_number = register_chrdev(0, DEVICE_NAME, &fops);
    if (major_number < 0) {
        printk(KERN_ALERT "LED Driver: Khong the dang ky Major number\n");
        return major_number;
    }

    // Tạo Class (Requirement 15)
    led_class = class_create(THIS_MODULE, CLASS_NAME);
    if (IS_ERR(led_class)) {
        unregister_chrdev(major_number, DEVICE_NAME);
        return PTR_ERR(led_class);
    }

    // Tạo Device file trong /dev (Requirement 12, 16)
    led_device = device_create(led_class, NULL, MKDEV(major_number, 0), NULL, DEVICE_NAME);
```

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^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next

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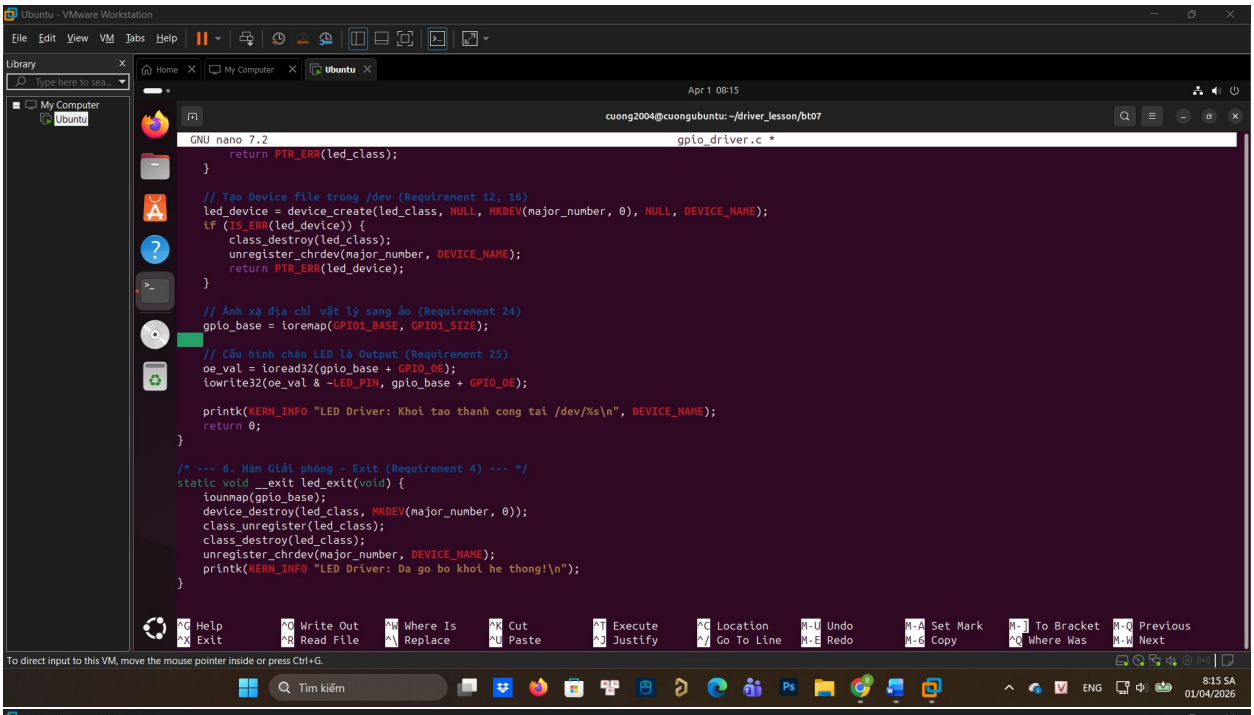

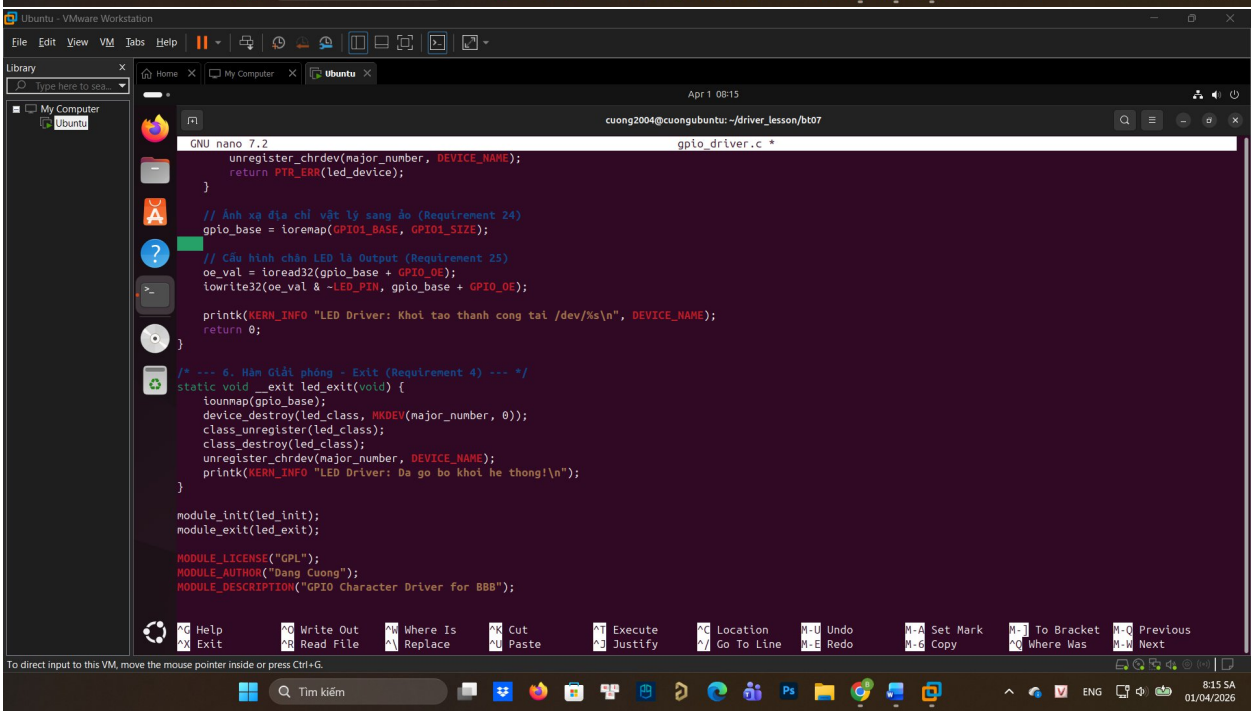

Viết file Makefile

```
cuong2004@cuongubuntu:~/driver_lesson/bt07$ nano Makefile
```

Biên dịch

```
cuong2004@cuongubuntu:~/driver_lesson/bt07$ make
make ARCH=arm CROSS_COMPILE=/home/cuong2004/buildroot/output/host/bin/arm-linux- -C /home/cuong2004/buildroot/output/build/linux-custom M=/home/cuong2004/driver_le
sson/bt07 modules
make[1]: Entering directory '/home/cuong2004/buildroot/output/build/linux-custom'
  CC [M]  /home/cuong2004/driver_lesson/bt07/gpio_driver.o
  MODPOST /home/cuong2004/driver_lesson/bt07/Module.symvers
  CC [M]  /home/cuong2004/driver_lesson/bt07/gpio_driver.mod.o
  LD [M]  /home/cuong2004/driver_lesson/bt07/gpio_driver.ko
make[1]: Leaving directory '/home/cuong2004/buildroot/output/build/linux-custom'
cuong2004@cuongubuntu:~/driver_lesson/bt07$
ouse pointer inside or press Ctrl+G.
```

Kết nối thẻ nhớ

```
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cuong2004@cuongubuntu: ~/driver_lesson/bt07
loop0    7:0    0     4K  1 loop /snap/bare/5
loop1    7:1    0 273.5M  1 loop /snap/firefox/8054
loop2    7:2    0    74M  1 loop /snap/core22/2339
loop3    7:3    0  66.8M  1 loop /snap/core24/1499
loop5    7:5    0 254.6M  1 loop /snap/firefox/7901
loop6    7:6    0  18.5M  1 loop /snap/firmware-updater/212
loop7    7:7    0  16.4M  1 loop /snap/firmware-updater/216
loop8    7:8    0   516M  1 loop /snap/gnome-42-2204/202
loop9    7:9    0 606.1M  1 loop /snap/gnome-46-2404/153
loop10   7:10   0  91.7M  1 loop /snap/gtk-common-themes/1535
loop11   7:11   0   395M  1 loop /snap/mesa-2404/1165
loop12   7:12   0  10.8M  1 loop /snap/snap-store/1270
loop13   7:13   0  15.5M  1 loop /snap/snap-store/1310
loop14   7:14   0  48.1M  1 loop /snap/snapd/25935
loop15   7:15   0  48.4M  1 loop /snap/snapd/26382
loop16   7:16   0   576K  1 loop /snap/snapd-desktop-integration/315
loop17   7:17   0   576K  1 loop /snap/snapd-desktop-integration/343
loop18   7:18   0    74M  1 loop /snap/core22/2411
loop19   7:19   0  66.8M  1 loop /snap/core24/1587
sda      8:0    0    50G  0 disk
├sda1   8:1    0     1M  0 part
└sda2   8:2    0    50G  0 part /
sdb      8:16   1  29.7G  0 disk
├sdb1   8:17   1    16M  0 part /media/cuong2004/80F2-D8CA
└sdb2   8:18   1   512M  0 part /media/cuong2004/rootfs
sr0     11:0    1   5.9G  0 rom  /media/cuong2004/Ubuntu 24.04.3 LTS amd64
cuong2004@cuongubuntu:~/driver_lesson/bt07$ sudo cp ~/driver_lesson/bt07/gpio_driver.ko /media/cuong2004/rootfs/home/debian/
[sudo] password for cuong2004:
cp: cannot create regular file '/media/cuong2004/rootfs/home/debian/': No such file or directory
cuong2004@cuongubuntu:~/driver_lesson/bt07$ # Cách 1: Copy vào thư mục root của hệ điều hành trên thẻ
sudo cp ~/driver_lesson/bt07/gpio_driver.ko /media/cuong2004/rootfs/root/

# Cách 2 (Chắc chắn nhất): Nếu lệnh trên vẫn báo lỗi, hãy copy vào ngay ngoài cùng
sudo cp ~/driver_lesson/bt07/gpio_driver.ko /media/cuong2004/rootfs/
cuong2004@cuongubuntu:~/driver_lesson/bt07$ sync
cuong2004@cuongubuntu:~/driver_lesson/bt07$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Tìm kiếm
ENG
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```

Vào BBB

Tìm và nạp Driver

```
Welcome to Buildroot
buildroot login: root
# insmod gpio_driver.ko
[   16.266015] gpio_driver: loading out-of-tree module taints kernel.
[   16.274019] LED Driver: Dang bat dau khoi tao...
[   16.282480] LED Driver: Khoi tao thanh cong tai /dev/my_gpio_led
#
```

Kiểm tra Device File

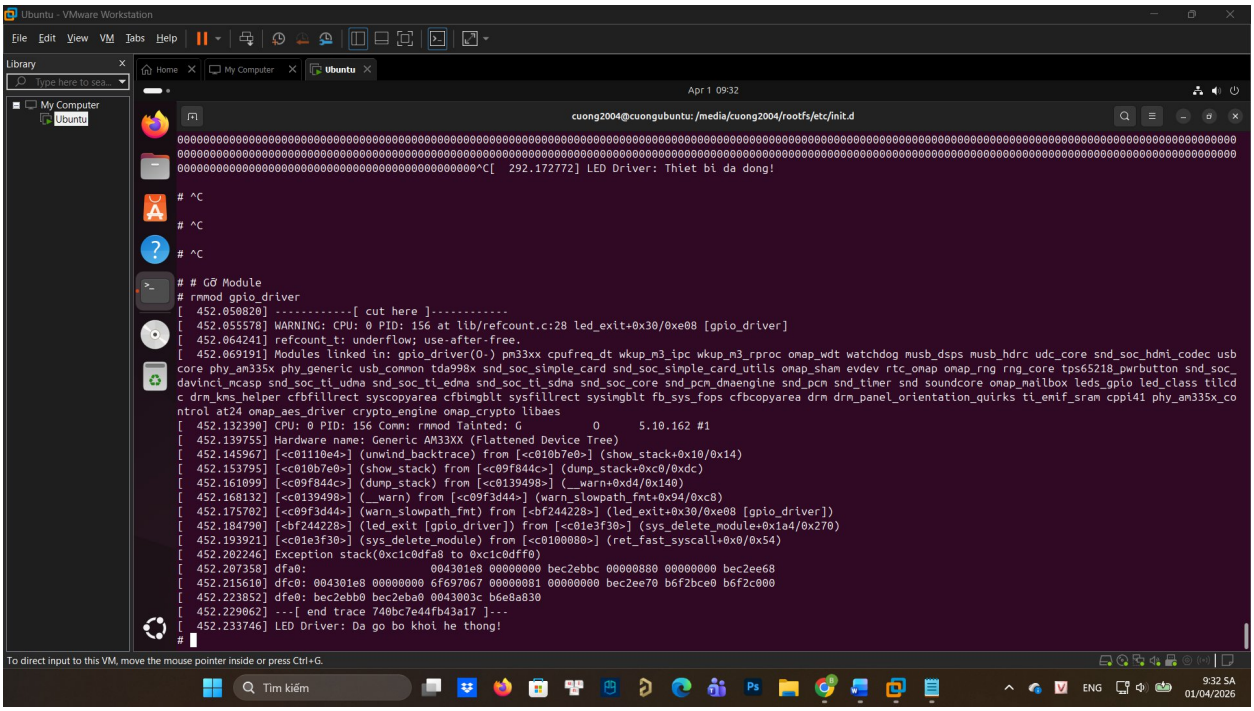
```
# ls -l /dev/my_gpio_led
crw-rw----    1 root     root      246,   0 Jan  1 00:00 /dev/my_gpio_led
#
```

Test chức năng Ghi (Write) & ioremap

```
# echo "1" > /dev/my_gpio_led
[  130.514891] LED Driver: Thiet bi da mo!
[  130.519514] LED Driver: LED ON
[  130.522667] LED Driver: Thiet bi da dong!
# echo "0" > /dev/my_gpio_led
[  141.085556] LED Driver: Thiet bi da mo!
[  141.089924] LED Driver: LED OFF
[  141.093160] LED Driver: Thiet bi da dong!
#
```

Test chức năng Đọc (Read)

```
0000000000
```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cuong2004@cuongubuntu: /media/cuong2004/rootfs/etc/init.d
0000000000

Test open/release

```
# echo "1" > /dev/my_gpio_led
[  847.555651] LED Driver: Thiet bi da mo!
[  847.560071] LED Driver: LED ON
[  847.563220] LED Driver: Thiet bi da dong!
```

các lệnh Shell cơ bản như echo hoặc cat sẽ thực hiện quy trình: Mở file (Open) -> Thao tác (Read/Write) -> Đóng file (Release).

Test copy to user

```
# cat /dev/my_gpio_led
1
```